

FoilDSL v3 — Language Specification

Sep 22, 2026 · @Someone

1. Status and scope

FoilDSL v3 is a human-readable language that defines one lifting surface of a hydrofoil — a front wing, a stabilizer or a fin — as a deterministic function of its source text. This document is the normative definition of version 3; the FoilDSL Explorer v3 is its reference implementation.

A program describes five things: the planform (leading and trailing edges), the out-of-plane shape (drop and rise from the center line), the spanwise twist, a schedule of sections, and the loft sampling. It may also carry constraints, which are checked against the projected geometry but never change it.

Out of scope for v3: multi-surface assemblies (mast, fuselage, stabilizer placement), trailing-edge thickness, and loading or hydrodynamic prediction. Those belong to later versions or to the solver.

A **conforming parser** accepts exactly the programs generated by the grammar in section 13 that also satisfy every rule in section 10, and rejects all others with a line number and a reason. A **conforming projector** maps an accepted program to the surface defined in section 11, and to the same surface every time.

2. Notation and conventions

MUST, MUST NOT and SHOULD carry their RFC 2119 meanings. Grammar rules use EBNF: terminals in quotes, `{ x }` for zero or more, `[ x ]` for optional, `|` for alternation, `(* ... *)` for comments.

The coordinate frame is fixed:

Axis	Symbol	Direction	Origin
Streamwise	x	aft, with the flow	leading edge at the center line
Spanwise	y	outward to the tip	center line
Vertical	z	toward the board (up)	center-line leading edge height

Only the half span $0 \leq y \leq b/2$ is written; the other half is its mirror. All lengths are millimetres after resolution; a length may be written in cm or m and is converted at parse time. Angles are degrees in source and radians inside the projection. Positive twist is nose-up, so washout is negative.

3. Lexical structure

Source is UTF-8 text read as a sequence of tokens. Whitespace separates tokens and is otherwise ignored; line breaks are counted for error reporting only. A `#` starts a comment that runs to the end of the line.

Token	Form	Examples
word	letter or <code>_</code> , then letters, digits, <code>_</code>	<code>span</code> , <code>monotone</code> , <code>E817</code> , <code>cm2</code>
number	optional sign, digits, optional fraction	<code>900</code> , <code>-2</code> , <code>+46</code> , <code>0.197</code> , <code>.5</code>
string	double-quoted, single line	<code>"gull-90"</code>
relation	<code>>=</code> <code><=</code> <code>==</code> <code>~</code>	
symbol	<code>{ }</code> <code>()</code> <code>[]</code> <code>:</code> <code>,</code> <code>%</code>	

Keywords are ordinary words recognised by position, so a keyword may still appear as a name where no keyword is expected. A character that starts no token is a lexical error at its line.

Two lexical consequences matter to authors. A NACA designation such as `0010` is a number token, not a word. In `naca 0012-64` the modifier lexes as the signed number `-64`, which the section rule reads as a modifier only when it has exactly two digits.

4. Program structure

A program is exactly one `foil "<name>" { ... }` block; any token after its closing brace is an error. Statements inside it may appear in any order, and a later statement of the same kind replaces an earlier one.

Statement	Required	Meaning
<code>units mm</code>	no	declares millimetres; the only accepted value
<code>span <length></code>	yes	tip-to-tip span b
<code>context { <name> <number> <unit> ... }</code>	no	annotation (rider mass, board); ignored by the projection
<code>planform { ... }</code>	yes	leading and trailing edge curves, or a chord law
<code>dihedral <mode> { ... }</code>	no	out-of-plane curve $z(y)$; absent means flat
<code>twist <mode> { ... }</code>	no	twist curve $\phi(y)$ in degrees; absent means zero
<code>sections [linear cosine] { ... }</code>	yes	the section schedule
<code>loft { ... }</code>	no	sampling; defaults to degree 3, 21×81 cosine
<code>constrain { ... }</code>	no	assertions checked against the projection

Station references are resolved after the whole block is read, so `span` need not come before the curves that depend on it.

5. Stations and curves

Every spanwise quantity is a curve: an ordered list of anchors `at <station> <value>` plus an interpolation mode. A station is a distance from the center line.

Station form	Resolves to
<code>center</code> (alias <code>root</code>)	$y = 0$
<code>tip</code>	$y = b/2$
<code><number> [mm cm m]</code>	that distance, in mm
<code><number> %</code>	that fraction of the half span

A curve MUST start at `center`, end at `tip`, and list stations strictly outward. It MUST have at least two anchors. Values outside $[0, b/2]$ are unreachable by construction.

With anchors (y_i, v_i) , slopes m_i and $h_i = y_{i+1} - y_i$, every non-linear mode evaluates the same cubic Hermite form on each interval, with $s = (y - y_i)/h_i$:

$$f(y) = (2s^3 - 3s^2 + 1) \setminus v_i + (s^3 - 2s^2 + s) \setminus h_i m_i + (-2s^3 + 3s^2) \setminus v_{i+1} + (s^3 - s^2) \setminus h_i m_{i+1}$$

The modes differ only in how the slopes are chosen:

Mode	Slopes m_i	Continuity	Overshoot
<code>linear</code>	none; straight segments	C^0 , corners at anchors	never
<code>smooth</code>	Catmull-Rom: $(v_{i+1} - v_{i-1})/(y_{i+1} - y_{i-1})$; end slopes are the end secants	C^1	possible
<code>monotone</code>	Fritsch-Carlson: mean of adjacent secants, 0 where they change sign, limited so $\alpha^2 + \beta^2 \leq 9$	C^1	never

The consequence authors rely on: in `monotone` mode every local extreme of the curve lies on an anchor, so a written `drop 16 mm` is exactly the deepest point. Outside $[0, b/2]$ a curve holds its end value.

6. Planform

The planform is two independent edge curves, `leading` and `trailing`, each giving streamwise position x (mm aft of the center-line leading edge) against y . Because the edges are separate, every sweep permutation is expressible: leading edge aft with trailing edge forward (a crescent), both aft, both forward, or an edge that reverses along the span.

```
planform {
  leading smooth {
    at center    x 0
    at 225 mm    x 14
    at tip       x 96
  }
  trailing monotone {
    at center    x 132
    at 225 mm    x 116
    at tip       x 122
  }
}
```

The `x` keyword before a value is optional. Everything else is derived:

Quantity	Definition
chord	$c(y) = x_{TE}(y) - x_{LE}(y)$
quarter-chord spine	$x_q(y) = x_{LE}(y) + \frac{1}{4} \cdot c(y)$
center chord	$c_0 = c(0)$
tip chord	$c_{tip} = c(b/2)$
taper	$\lambda = c_{tip} / c_0$
local sweep	$\Lambda(y) = \text{atan } x'(y)$, per edge, + aft

Taper is never authored; it is read off the two edges and may be constrained.

A chord law is shorthand that expands to edge curves about a straight quarter-chord line:

`chord elliptic center 132 tip 26` or `chord linear center 132 taper 0.2`, optionally followed by `sweep 3 deg`. The elliptic law $c(y) = c_{tip} + (c_0 - c_{tip}) \cdot \sqrt{1 -$

$(2y/b)^2$) is sampled at 0, 30, 55, 75, 88, 96 and 100% of the half span with `smooth` edges; the linear law uses two `linear` anchors. If both a chord law and explicit edges are given, the explicit edges win.

7. Out-of-plane shape

The `dihedral` curve gives the vertical offset $z(y)$ of the reference line, measured from 0 at the center. Positive z is toward the board. Each anchor names a distance from the center and a drop or rise:

Value form	z
<code>level</code>	0
<code>drop <length></code>	-length
<code>rise <length></code> (alias <code>raise</code>)	+length
<code>[z] <length></code>	signed length as written

```
dihedral monotone {
  at center    level
  at 135 mm    drop 16 mm      # bottoms out, turns back up
  at 270 mm    level          # back through 0
  at tip       rise 46 mm      # tip lift
  bank true
}
```

`bank true` (the default) rotates each section about the streamwise axis by the local dihedral $\Gamma(y) = \text{atan } z'(y)$, so section planes stay normal to the curve; `bank false` keeps them vertical.

A conforming tool SHOULD report the out-of-plane features below, each as a distance from the center and a magnitude from 0. These are computed, not authored, so they hold for any mode.

Feature	Condition	Reported as
low (bottoms out)	z' changes $- \rightarrow +$	y, z (a drop if $z < 0$)
high (peaks)	z' changes $+ \rightarrow -$	y, z

Feature	Condition	Reported as
return through 0	z changes sign	y, heading up or down
inflection	z'' changes sign	y, u bowl ↔ n crest
stretch	contiguous sign of z'	y range, rising (dihedral) / falling (anhedral) / level, Γ range
max drop / max rise	extremes of z on [0, b/2]	magnitude and y

In `linear` mode z'' is zero between anchors; interior anchors are reported as kinks instead of inflections.

8. Sections

A section is four parameters on a unit chord: thickness t, the chordwise position x_t where thickness peaks, camber m, and the position x_m where camber peaks. The `sections` block places sections at stations; between stations every parameter blends.

```
sections cosine {
  at center      naca 4412
  at 250 mm      profile { thickness 10% at 33%  camber 2% at 40% }
  at tip         profile { thickness 8% at 30%  camber 0% }
}
```

Source	t	x_t	m	x_m
naca MPTT	TT / 100	30%	M / 100	P / 10 (40% if P = 0)
naca MPTT-IT	TT / 100	T / 10	M / 100	P / 10
profile { thickness t [at x_t] camber m [at x_m] }	t	x_t (default 30%)	m (default 0)	x_m (default 40%)
eppler <name>, file "<path>"	nominal 10%	30%	2%	40%

Percentages need a % sign; a bare value is read as a fraction and MUST NOT exceed 1.

`family naca` (default) or `family cst` in a profile selects the base thickness shape.

External sections are nominal until their coordinates are resolved by a later stage; tools SHOULD flag them.

Thickness is placed by stretching a base shape Φ , whose own maximum sits at u^* , so the maximum lands exactly at x_t :

$$y_t(\xi) = \frac{t}{2} \frac{\Phi(u(\xi))}{\Phi(u^*)}, \quad \text{quad } u(\xi) = \begin{cases} u^* \xi / x_t & \xi \leq x_t \\ u^* + (1 - u^*) \frac{\xi - x_t}{1 - x_t} & \xi > x_t \end{cases}$$

$\Phi'(u^*) = 0$, so the stretched profile stays C^1 at x_t . For NACA, $\Phi(u) = 0.2969u - 0.1260u - 0.3516u^2 + 0.2843u^3 - 0.1036u^4$ (closed trailing edge). A side effect authors should expect: moving x_t aft sharpens the leading edge. Camber is the NACA 4-digit mean line with maximum m at x_m ; thickness is applied normal to it.

Between stations i and $i+1$, with $s = (y - y_i)/(y_{i+1} - y_i)$, each of (t, x_t, m, x_m) is $\sigma_i + w(s)(\sigma_{i+1} - \sigma_i)$, where $w(s) = s$ for `linear` and $w(s) = \frac{1}{2}(1 - \cos \pi s)$ for `cosine` (the default).

When one side has zero camber, its x_m is taken from the other side so the camber peak does not jump. The base family switches at $s = \frac{1}{2}$.

9. Twist, loft and constraints

Twist is a curve of incidence in degrees, `twist linear { at center 0 deg at tip -2 deg }`; the `deg` suffix is optional. Sections rotate about their quarter-chord point.

Loft fixes sampling: `surface bspline degree d, stations N uniform|cosine` (spanwise master curves) and `points M uniform|cosine` (chordwise samples per section). Defaults are degree 3, 21 stations and 81 points, all cosine-spaced.

Constraints are `metric relation value [unit]`. They are evaluated against the projection and never alter it. `~` holds within 5% of the value; `==` within 10^{-6} relative.

Metric	Dimension	Default unit	Definition
<code>area</code>	area	cm2	projected $S = 2 \int c \, dy$
<code>dev_area</code>	area	cm2	developed $2 \int c \cdot V(1 + z'^2) \, dy$
<code>aspect</code>	none	—	b^2 / S

Metric	Dimension	Default unit	Definition
<code>taper</code>	none	—	c_{tip} / c_0
<code>mac</code>	length	mm	$(2/S) \int c^2 dy$
<code>center_chord</code>	length	mm	c_0
<code>tip_chord</code>	length	mm	c_{tip}
<code>tip_rise</code>	length	mm	$z(b/2)$, signed
<code>max_drop</code>	length	mm	$\max(0, -\min z)$
<code>max_rise</code>	length	mm	$\max(0, \max z)$
<code>root_thick</code>	length	mm	$c_0 \cdot t(0)$
<code>tip_thick</code>	length	mm	$c_{\text{tip}} \cdot t(b/2)$

Length metrics accept mm, cm or m; area metrics accept mm2, cm2 or m2; dimensionless metrics MUST carry no unit.

10. Static semantics

A program that parses is accepted only if it satisfies every rule below. A conforming parser MUST reject a violation with the line of the offending token and a message naming the rule. Rules are checked in this order, and the first failure is reported.

#	Rule	Reported at
S1	<code>span</code> is present and positive	<code>foil</code> / <code>span</code>
S2	the planform has a leading and a trailing edge, explicitly or from a chord law	<code>planform</code>
S3	every curve starts at <code>center</code> , ends at <code>tip</code> , and its stations strictly increase	the offending station
S4	every curve has at least two anchors and a mode of <code>linear</code> , <code>smooth</code> or <code>monotone</code>	the mode
S5	the trailing edge stays at least 0.5 mm aft of the leading edge at 91 evenly spaced stations	<code>planform</code>

#	Rule	Reported at
S6	a sections block exists with at least one station; the first is at <code>center</code> , the last (if more than one) at <code>tip</code> , stations strictly increasing	the offending station
S7	every section has $1\% \leq t \leq 35\%$ and $15\% \leq x_t \leq 65\%$	the station
S8	every section has $0\% \leq m \leq 9\%$, and $15\% \leq x_m \leq 85\%$ when $m > 0$	the station
S9	a value written without <code>%</code> in a percentage position does not exceed 1	the value
S10	a NACA designation has four digits; a modifier has two	the designation
S11	<code>bank</code> is <code>true</code> or <code>false</code>	the value
S12	every constraint names a known metric, a relation, and a unit of the metric's dimension	the metric
S13	$1 \leq \text{loft degree} < \text{loft stations}$	<code>foil</code>
S14	<code>units</code> , if present, is <code>mm</code>	<code>units</code>

S5 samples rather than proves positivity; a crossing narrower than 5 mm can escape it. Tools SHOULD also warn, without rejecting, when a `smooth` curve overshoots its anchors.

11. Denotational semantics

The meaning of an accepted program F is a surface $\llbracket F \rrbracket \subset \mathbb{R}^3$, built by pure functions in the order below. Nothing depends on evaluation order, clock or randomness.

```

flowchart LR
  A[source] -->|parse| B[AST]
  B -->|resolve stations<br/>expand sugar| C[canonical model]
  C -->|curves + sections| D[placed sections]
  D -->|loft| E[B-spline surface]
  E -->|emit| F[STEP / grid]
```

The reference point of the station at y rides the quarter-chord spine and the out-of-plane curve, and the local bank angle comes from the slope of z :

$$r(y) = \big(x_{LE}(y) + \tfrac{1}{4} c(y), y, z(y)\big), \quad \Gamma(y) = \arctan z'(y)$$

A point (ξ, ζ) of the unit section $\sigma(y)$ is scaled by chord, rotated by twist ϕ about the spanwise axis, banked by Γ about the streamwise axis, and translated:

$$P(\xi, \zeta; y) = r(y) + c(y) \begin{bmatrix} D(\Gamma(y)) \\ R(\phi(y)) \end{bmatrix} \begin{bmatrix} \xi - \tfrac{1}{4} \\ \hat{e}_x + \zeta \hat{e}_z \end{bmatrix}$$

For the port side, y is replaced by $-y$ and the bank direction mirrored. The loft evaluates P at N spanwise and M chordwise stations (uniform, or cosine: $\eta_i = \frac{1}{2}(1 - \cos(i\pi/(N-1)))$) and fits a B-spline surface of the declared degree through that net.

Derived metrics (section 9) and reported features (section 7) are functions of the same canonical model, so they are equally deterministic.

12. Canonical form and identity

Every accepted program has one canonical text, $\text{canon}(F)$, and its identity is the hash of that text:

$$\text{id}(F) = H\big(\text{canon}(\text{parse}(s))\big)$$

The canonical emitter MUST produce:

1. statements in the fixed order: `units`, `span`, `context`, `planform`, `dihedral`, `twist`, `sections`, `loft`, `constrain`;
2. explicit `leading` and `trailing` curves, with chord-law sugar already expanded;
3. stations as `center`, `tip` or `<n> mm`, lengths rounded to 0.1 mm, percentages to 0.1%;
4. out-of-plane values as `level`, `drop <n> mm` or `rise <n> mm`;
5. a NACA station as its designation if untouched, otherwise as a `profile`;
6. no comments, two-space indentation, one anchor per line.

The emitter MUST be idempotent: $\text{canon}(\text{parse}(\text{canon}(F))) = \text{canon}(F)$. Comments, spacing and statement order in the source therefore never change $\text{id}(F)$; any change to geometry, sections, sampling or constraints does. The reference implementation uses 32-bit FNV-1a for H ; a production system SHOULD use SHA-256 and MAY truncate it for display.

An interactive editor that lets people shape the foil directly SHOULD write every edit back through the canonical emitter and re-check it against section 10, so the program stays the single source of truth.

13. Complete grammar

The grammar is LL(1): every choice is decided by the next token, so a recursive-descent parser needs no backtracking. Lexical classes `word`, `number` and `string` are defined in section 3.

```

program      = "foil" string "{" { statement } "}" ;

statement    = "units" "mm"
              | "span" length
              | "context" "{" { word number word } "}"
              | planform
              | "dihedral" zcurve
              | "twist" degcurve
              | sections
              | loft
              | constrain ;

(* ---- stations, lengths, percentages ---- *)
station      = "center" | "root" | "tip" | number [ lenunit | "%" ] ;
length       = number [ lenunit ] ;
lenunit      = "mm" | "cm" | "m" ;
pct          = number [ "%" ] ;                (* without % : fraction <= 1 *)

(* ---- curves ---- *)
mode         = "linear" | "smooth" | "monotone" ;
xcurve       = mode "{" { "at" station [ "x" ] length } "}" ;
zcurve       = mode "{" { "at" station zvalue | "bank" ( "true" | "false" ) }
              "}" ;
zvalue       = "level" | "drop" length | ( "rise" | "raise" ) length | [ "z" ]
              length ;
degcurve     = mode "{" { "at" station number [ "deg" ] } "}" ;

(* ---- planform ---- *)
planform     = "planform" "{" { planstmt } "}" ;
planstmt     = "leading" xcurve
              | "trailing" xcurve
              | "chord" ( "elliptic" | "linear" ) ( "center" | "root" ) length

```

```

        ( "tip" length | "taper" number ) [ "sweep" number "deg" ] ;

(* ---- sections ---- *)
sections      = "sections" [ "linear" | "cosine" ] "{" { "at" station source }
               "}" ;
source        = "naca" number [ number ]      (* 4 digits, optional -IT
modifier *)
               | "profile" "{" { profstmt } "}"
               | "eppler" word
               | "file" string ;
profstmt      = "family" ( "naca" | "cst" )
               | "thickness" pct [ "at" pct ]
               | "camber" pct [ "at" pct ] ;

(* ---- loft ---- *)
loft          = "loft" "{" { loftstmt } "}" ;
loftstmt      = "surface" "bspline" "degree" number
               | ( "stations" | "points" ) number ( "uniform" | "cosine" ) ;

(* ---- constraints ---- *)
constrain     = "constrain" "{" { metric relation number [ unit ] } "}" ;
metric        = "area" | "dev_area" | "aspect" | "taper" | "mac"
               | "center_chord" | "tip_chord" | "tip_rise"
               | "max_drop" | "max_rise" | "root_thick" | "tip_thick" ;
relation      = ">=" | "<=" | "==" | "~" ;
unit          = lenunit | "mm2" | "cm2" | "m2" ;

```

A `profile` MUST contain a `thickness` statement (S7 then bounds it).

14. Reference program

`gull-90` is the conformance reference: a 900 mm crescent front wing that drops 16 mm at 135 mm from center, returns through 0 at 270 mm and lifts 46 mm at the tip. A conforming implementation MUST reproduce the values below within 0.5% for integrated quantities and 1 mm for feature positions.

```

foil "gull-90" {
  units mm
  span 900 mm
  context { rider_mass 90 kg }

  planform {

```

```

leading smooth {
  at center    x 0
  at 225 mm    x 14
  at tip       x 96
}
trailing monotone {
  at center    x 132
  at 225 mm    x 116
  at tip       x 122
}
}

dihedral monotone {
  at center    level
  at 135 mm    drop 16 mm
  at 270 mm    level
  at tip       rise 46 mm
  bank true
}

twist linear {
  at center    0 deg
  at tip       -2 deg
}

sections cosine {
  at center    naca 4412
  at 250 mm    profile { thickness 10% at 33%  camber 2% at 40% }
  at tip       profile { thickness 8% at 30%  camber 0% }
}

loft {
  surface  bspline degree 3
  stations 21 cosine
  points   81 cosine
}

constrain {
  area      >= 740 cm2
  aspect    ~ 9.5
  taper     <= 0.25
  max_drop  <= 20 mm
}

```

```

    tip_rise  >= 30 mm
    tip_thick >= 1.2 mm
  }
}

```

Quantity	Expected
center chord / tip chord	132.0 mm / 26.0 mm
taper λ	0.197
projected area / developed area	832 cm ² / 843 cm ²
aspect ratio / MAC	9.74 / 103 mm
low (max drop)	16.0 mm below 0 at 136 mm
return through 0	270 mm, heading up
inflections	45 mm ($n \rightarrow u$), 390 mm ($u \rightarrow n$)
stretches	falling 0–137 mm ($\Gamma -9.0^\circ \dots -0.4^\circ$); rising 137–450 mm ($\Gamma 0.3^\circ \dots 15.6^\circ$)
section at 125 mm	t 11.0% at 31.5%, camber 3.0%
tip thickness	2.1 mm
constraints	6 of 6 satisfied
id(F), FNV-1a of canon(F)	d801dac9

Negative tests: each edit to the program above **MUST** be rejected under the rule shown, at the line of the edited token.

Edit	Rule	Message (reference implementation)
thickness 10% at 33% → thickness 10 at 33%	S9	thickness 10 needs a % sign
at 33% → at 80%	S7	max-thickness position 80% is outside 15%–65% of chord
tip x 122 → x 60	S5	the trailing edge meets or crosses the leading edge 395 mm from center

Edit	Rule	Message (reference implementation)
<code>at 270 mm level</code> → <code>at 100 mm level</code>	S3	stations in the dihedral curve must move outward
<code>dihedral monotone</code> → <code>dihedral wiggly</code>	S4	unknown curve mode "wiggly"
<code>max_drop <= 20 mm</code> → <code><= 20 cm2</code>	S12	max_drop is a length; "cm2" is not a length unit

15. Changes from v1/v2 and open issues

Area	v1	v2	v3
spanwise coordinate	fraction η	fraction η	distance from center, mm
planform	chord law + one sweep angle	LE/TE curves	LE/TE curves; chord law is sugar
out-of-plane	one dihedral angle	$z(\eta)$ curve	drop / rise / level from 0, with reported turning points
sections	family + thickness	family + thickness	thickness, max-thickness position, camber, camber position
section schedule	3 stations, blend	2 stations	any number of stations at mm positions
taper	authored	derived	derived as c_{tip} / c_o , constrainable
identity	hash of AST	hash of AST	hash of canonical text; emitter is idempotent

v3 is not source-compatible with v1: `planform { span ... }` moved to top-level `span`, and `twist { law washout ... }` became a curve.

Open issues for v4:

- ☐ Trailing-edge thickness, needed for manufacturable foils, as a fifth section parameter.

- ☐ A leading-edge radius parameter independent of x_t , as the NACA modifier's first digit implies.
- ☐ Resolving `eppler` and `file` sections from coordinates instead of nominal parameters.
- ☐ Multi-surface programs: mast, fuselage and stabilizer placed relative to the front wing.
- ☐ Replacing the sampled chord-positivity check (S5) with an exact test on the spline pieces.